Modern Physics

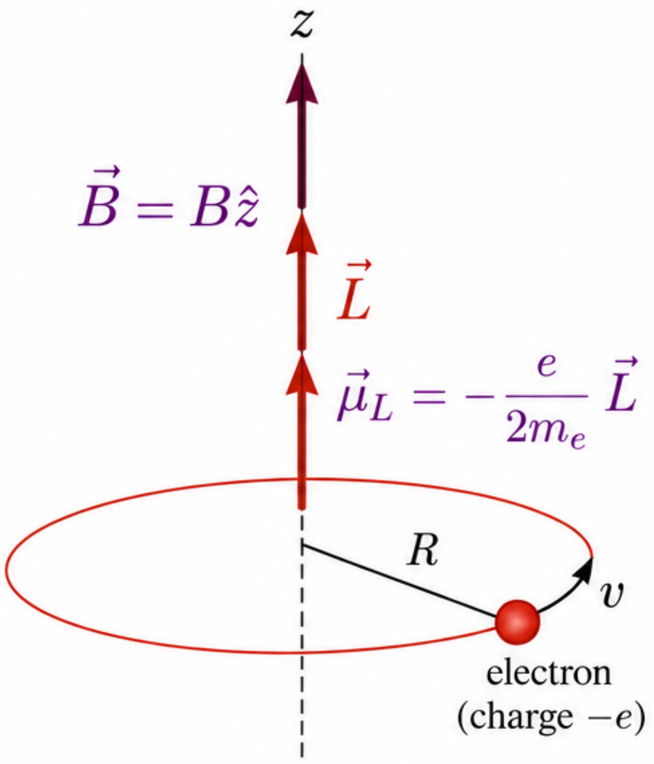
Lecture 6: Pauli Exclusion, Spin Orbit Coupling and X-ray spectra

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Books: *Ch 7 of* Arthur Beiser,
Concepts of Modern Physics

Atom in a uniform external magnetic field: Zeeman effect (orbital)

Orbital magnetic moment in a uniform magnetic field



Orbital magnetic moment:

$$\vec{\mu}_L = -\frac{e}{2m_e}\vec{L}.$$

1 Interaction with an external magnetic field

The interaction energy is

$$U = -\vec{\mu}_L \cdot \vec{B} = \frac{e}{2m_e}\vec{L}\cdot\vec{B}.$$

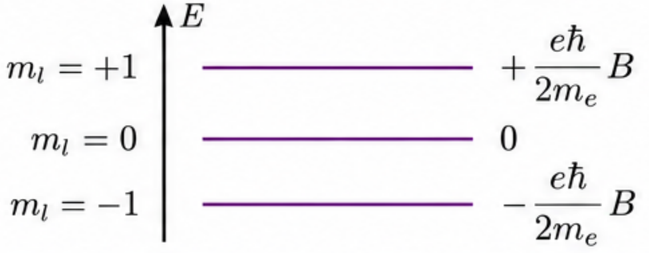
2 Energy shifts

For $\vec{B} = B\hat{z}$ and $L_z = m_l\hbar$ (with $m_l = -\ell, -\ell+1, \ldots, \ell$),

$$E_{ml} = \frac{e\hbar}{2m_e} B\, m_l, \qquad m_l = -\ell, -\ell+1, \ldots, \ell.$$

3 Zeeman splitting

The $(2\ell+1)$ degenerate levels split linearly with m_l. For $\ell = 1$ (p state):



4 Force and torque

The force on a magnetic dipole is $\vec{F} = -\nabla U$.
For a uniform $\vec{B}$, U is spatially constant $\Rightarrow$

$$\vec{F} = 0$$

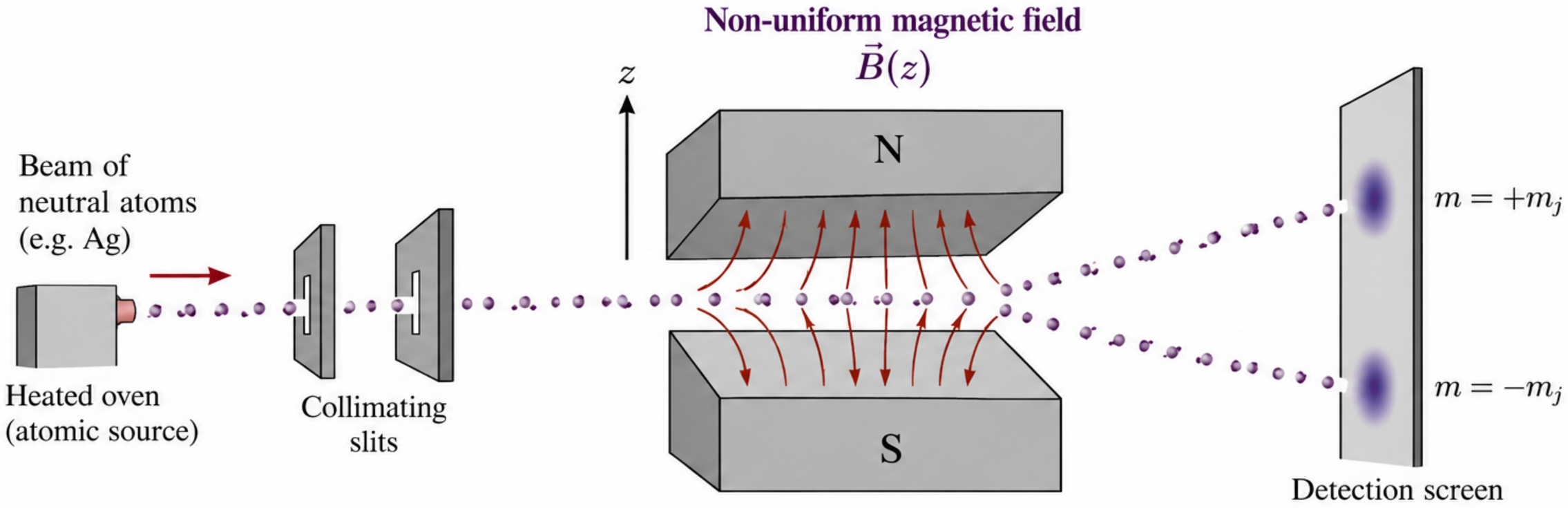
(no net translational force)

The torque is $\vec{\tau} = \vec{\mu}_L \times \vec{B}$, which is in general non–zero and tends to align $\vec{\mu}_L$ with $\vec{B}$.

$$\vec{\tau} \neq 0$$

(torque is non zero)

Stern Gerlach Experiment



Stern Gerlach experiment for l = 1

Non-uniform field and force

The potential energy of a magnetic dipole is

$$U = -\vec{\mu}_L \cdot \vec{B} = -\mu_{Lz} B(z).$$

The force along z is

$$F_z = -\frac{dU}{dz} = \mu_{Lz} \frac{dB}{dz}.$$

Since the field is stronger at the top, $\frac{dB}{dz} > 0$, **so the force is non-zero.**

Orbital magnetic moment and m_l

For an electron in an atomic orbital,

$$\vec{\mu}_L = -\frac{e}{2m_e}\vec{L}, \quad L_z = m_l \hbar,$$

so

$$\mu_{Lz} = -\frac{e\hbar}{2m_e} m_l.$$

$$F_z = -\frac{e\hbar}{2m_e} m_l \frac{dB}{dz}.$$

Deflection for $l = 1$ (three spots)

For $l = 1$, $m_l = -1, 0, +1$:

m_l	μ_{Lz}	F_z (for $dB/dz > 0$)	Deflection
$+1$	$-\frac{e\hbar}{2m_e}$	> 0	upward
0	0	0	no deflection
-1	$\frac{e\hbar}{2m_e}$	< 0	downward

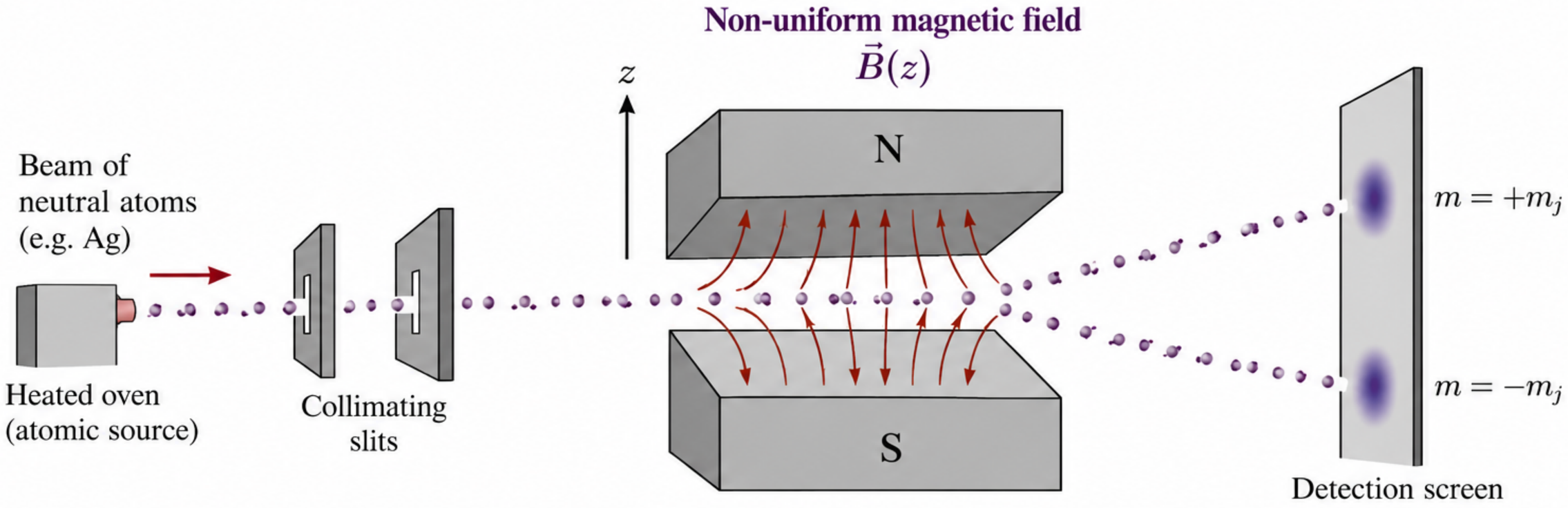
Thus, a beam of atoms with $l = 1$ is **expected to split into three spatially separated spots.**

The splitting we are talking about here is spitting seen on screen because some atoms are deflected upward and some downward. Splitting in Zeemen effect is in energy levels.

Stern Gerlach experiment for $l = 0$

If I send H atoms where electron are in ground state, that is orbital angular momentum is 0, we expect all atoms to be concentrate around centre.

Stern Gerlach Experiment



But we get two spots for l=0 electron state. Note that the spots are actually deflection of atom's trajectory. These must be due to the force we discussed two slides back.

Electron Spin

The electron possesses an additional, **intrinsic angular momentum S** called **spin**.

For an electron:

$$s = \frac{1}{2}, \qquad m_s = \pm\frac{1}{2}.$$

Hence the two Stern–Gerlach outcome.

It will also follow similar algebra:

$$S^2 = s(s+1)\hbar^2, \qquad s = 0,1,2,\ldots,n-1$$

$$S_z = m_s\hbar, \qquad m_s = -s, -s+1,\ldots,0,\ldots,s-1,s$$

Spin is a intrinsic property

Different quantum particles have different spins. Electrons are spin 1/2, photons are spin 1, Higgs spin 0.

Fermions and Bosons

Two families of quantum particles

Integer spin $0,1,2,\ldots \longrightarrow$ Bosons

Half-integer spin $\frac{1}{2}, \frac{3}{2}, \ldots \longrightarrow$ Fermions

Why should we care which family a particle belongs to?

Fermions - Pauli Exclusion Principle

For a free electron, suppose a convenient one-particle state is specified by its exact momentum and spin projection:

$$A = (\mathbf{p}, m_s), \qquad m_s = \pm\frac{1}{2}.$$

Can two free electrons have exactly the same state?

$(\mathbf{p}_1,+\frac{1}{2})$, $(\mathbf{p}_1,+\frac{1}{2})$ Not allowed

$(\mathbf{p}_1,+\frac{1}{2})$, $(\mathbf{p}_1,-\frac{1}{2})$ Allowed

$(\mathbf{p}_1,+\frac{1}{2})$, $(\mathbf{p}_2,+\frac{1}{2})$ Allowed

Position and momentum: exact momentum vs wave packets

1. Exact momentum state

If the momentum is exactly known,

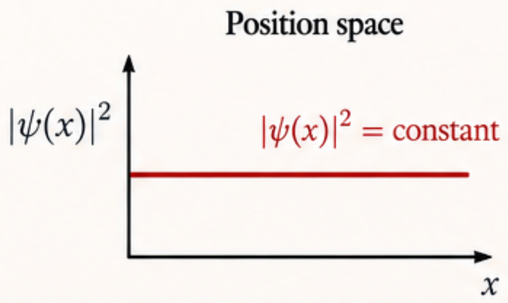
$$p = p_0 \quad (\Delta p = 0),$$

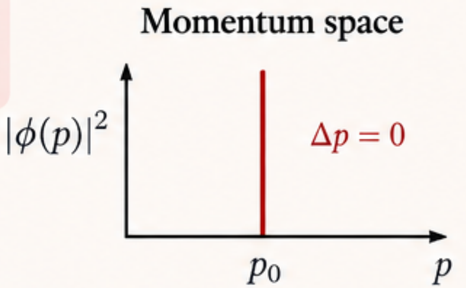
then from the uncertainty principle

$$\Delta x\, \Delta p \geq \frac{\hbar}{2} \quad \Rightarrow \quad \Delta x \to \infty.$$

A state with exact momentum is completely delocalized in position.

So we cannot assign two electrons to different locations with the same exact momentum.





2. Localized wave packet

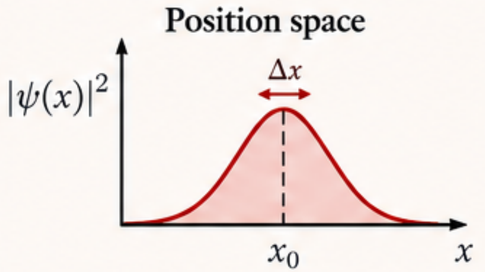
A localized state can be described by a Gaussian wave packet

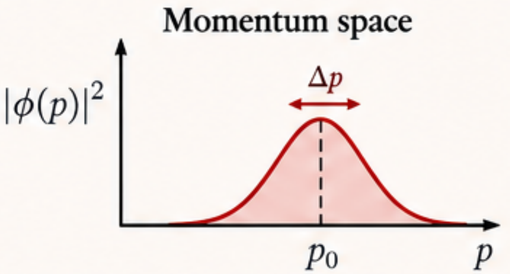
$$\psi(x) = \frac{1}{(2\pi\sigma_x^2)^{1/4}} e^{-(x-x_0)^2/4\sigma_x^2} e^{ip_0 x/\hbar}.$$

It has finite uncertainties

$$\Delta x \sim \sigma_x, \quad \Delta p \sim \frac{\hbar}{2\sigma_x}, \quad \Delta x\, \Delta p \geq \frac{\hbar}{2},$$

with average position $\langle x \rangle = x_0$ and average momentum $\langle p \rangle = p_0$.





3. Two electrons with the same average momentum but at different locations

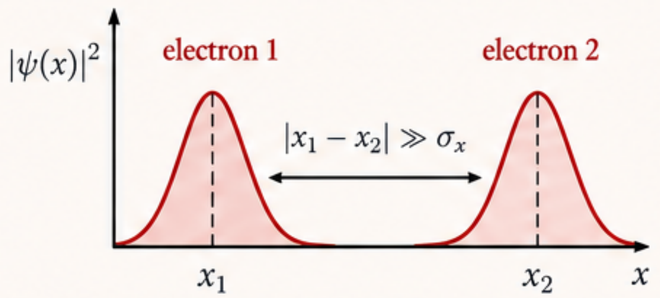
Take two Gaussian wave packets with the same average momentum (e.g. $\langle p \rangle = 0$) but centered at different positions x_1 and x_2.

$$\psi_1(x) = \frac{1}{(2\pi\sigma_x^2)^{1/4}} e^{-(x-x_1)^2/4\sigma_x^2}, \qquad \langle p \rangle_1 = 0,$$

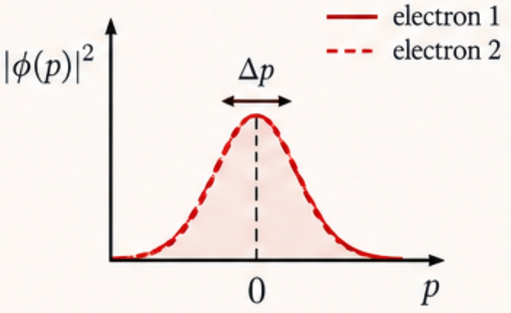
$$\psi_2(x) = \frac{1}{(2\pi\sigma_x^2)^{1/4}} e^{-(x-x_2)^2/4\sigma_x^2}, \qquad \langle p \rangle_2 = 0,$$

with $|x_1 - x_2| \gg \sigma_x$ (well separated).

Position space: two separated packets



Momentum space: same average momentum



Pauli Exclusion principle

For an electron in a hydrogen atom, a convenient one-particle state is specified by the quantum numbers n, l, m_l, m_s

$(n, l, m_l, +\frac{1}{2}), (n, l, m_l, +\frac{1}{2})$ **Not allowed**

$(n, l, m_l, +\frac{1}{2}), (n, l, m_l, -\frac{1}{2})$ **Allowed**

$(n, l, m_l, +\frac{1}{2}), (n', l', m_l', +\frac{1}{2})$ **Allowed if the spatial states are different**

For 1s orbital:

$(1,0,0, +\frac{1}{2}), (1,0,0, +\frac{1}{2})$ **Not allowed**

$(1,0,0, +\frac{1}{2}), (1,0,0, -\frac{1}{2})$ **Allowed**

Pauli Exclusion Principle

Two identical electrons occupy states A and B .

Possibility 1: $\psi_A(1)\psi_B(2)$

Possibility 2: $\psi_A(2)\psi_B(1)$

Are these physically different situations?

The electrons are identical --- we cannot tell which electron is 1 and which is 2.

So the wavefunction of the system must contain both possibilities:

$$\Psi(1,2) = \frac{1}{\sqrt{2}}\left[\psi_A(1)\psi_B(2) \pm \psi_A(2)\psi_B(1)\right]$$

Which sign should we choose for electrons?

Pauli Exclusion Principle

Pauli exclusion tells us the answer
Electrons cannot occupy the same complete quantum state.

So let $B = A$.

With the $+$ **sign:**

$$\Psi(1,2) = \sqrt{2}\,\psi_A(1)\psi_A(2) \neq 0$$

This would allow both electrons to occupy the same state.

With the $-$ **sign:**

$$\Psi(1,2) = \frac{1}{\sqrt{2}}\left[\psi_A(1)\psi_A(2) - \psi_A(2)\psi_A(1)\right] = 0$$

Therefore, for electrons:

$$\Psi(1,2) = -\,\Psi(2,1)$$

The two-electron wavefunction is antisymmetric under exchange.

Fermions $\longrightarrow$ antisymmetric wavefunction $\longrightarrow$ Pauli exclusion

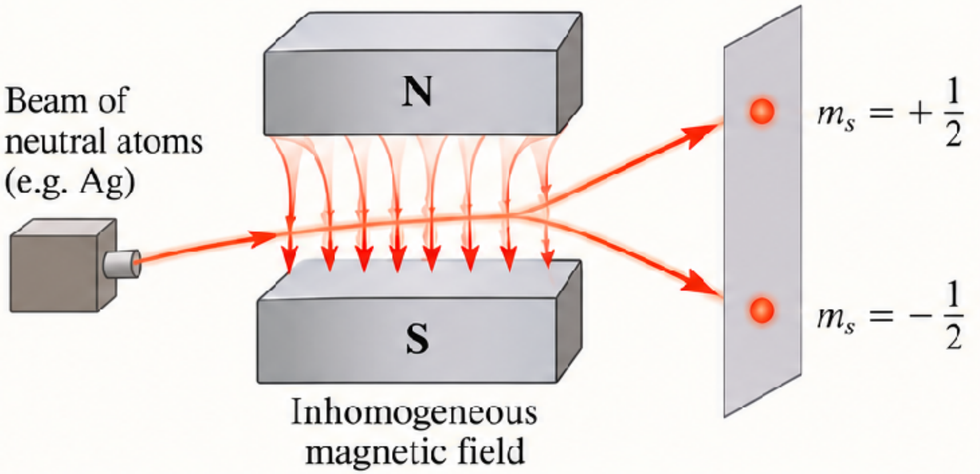
Bosons $\longrightarrow$ symmetric wavefunction

Pauli Exclusion Principle

Can we expect something similar to Zeeman effect but associated with Spin Angular momentum?

In principle, Yes.
We should

Stern–Gerlach experiment



Spin and magnetic moment

From the Stern–Gerlach experiment, we introduce an intrinsic angular momentum: **S**

Experiments show that spin obeys the same angular-momentum algebra as orbital angular momentum.

Orbital magnetic moment

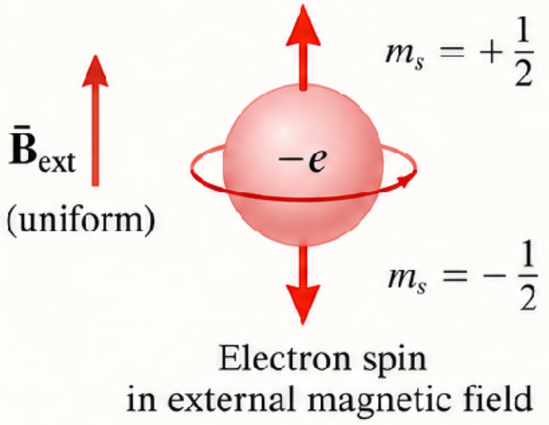
$$\boldsymbol{\mu}_L = -\frac{e}{2m_e}\mathbf{L}$$

Spin magnetic moment

$$\boldsymbol{\mu}_S = -g_s\frac{e}{2m_e}\mathbf{S}$$

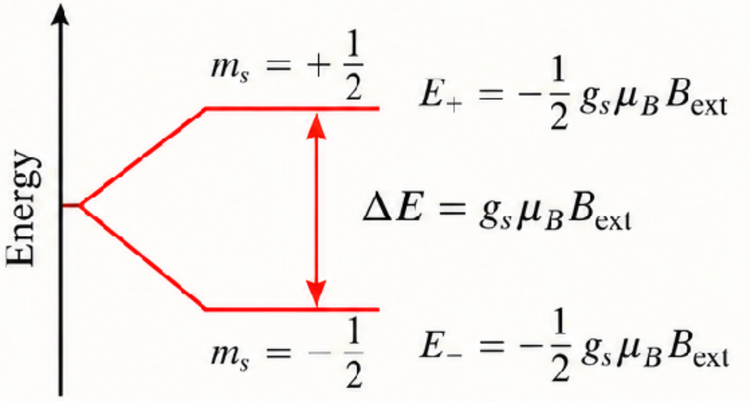
$$g_s \simeq 2$$

Spin contribution to the Zeeman effect



Interaction energy

$$\Delta H_{\text{spin}} = -\boldsymbol{\mu}_S \cdot \mathbf{B}_{\text{ext}} = g_s\frac{e}{2m_e}S_z B_{\text{ext}}$$



Spin-Orbit Interaction

But even when

$\mathbf{B}_{\text{ext}} = 0$

↓

the spin magnetic moment still experience a magnetic field

↓

Consider the electron's instantaneous rest frame

↓

Nucleus frame: electron moves in the electric field of the nucleus

↓

Electron rest frame: positively charged nucleus appears to move

↓

Moving charge produces a magnetic field $\mathbf{B}_{\text{eff}}$

1. Magnetic field of a moving charge

Start with the Biot–Savart law:

$$d\mathbf{B} = \frac{\mu_0}{4\pi}\,\frac{I\,d\boldsymbol{\ell} \times \mathbf{r}}{r^3}\,.$$

For a single moving charge, during a small time dt,

$$I = \frac{dq}{dt}, \qquad d\boldsymbol{\ell} = \mathbf{v}\,dt.$$

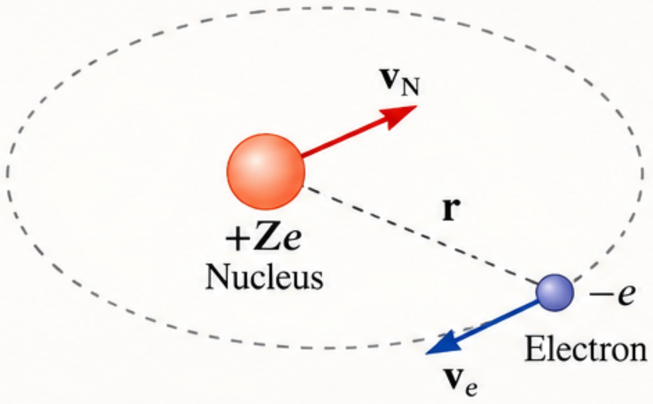
Therefore,

$$I\,d\boldsymbol{\ell} = \frac{dq}{dt}\,\mathbf{v}\,dt = dq\,\mathbf{v}.$$

For a point charge q,

$$\mathbf{B} = \frac{\mu_0}{4\pi}\,\frac{q\,\mathbf{v} \times \mathbf{r}}{r^3}\,.$$

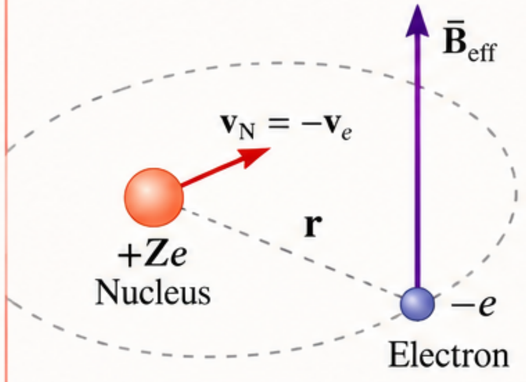
2. Nucleus viewed from the electron's rest frame



In the electron's instantaneous rest frame, the positively charged nucleus appears to move with

$$\mathbf{v}_N = -\mathbf{v}_e\,.$$

3. Magnetic field at the electron due to the moving nucleus



Using the result for a moving charge,

$$\mathbf{B}_{\text{eff}} = \frac{\mu_0}{4\pi}\,\frac{Ze\,\mathbf{v}_N \times \mathbf{r}}{r^3}\,.$$

With $\mathbf{v}_N = -\mathbf{v}_e$,

$$\mathbf{B}_{\text{eff}} = \frac{\mu_0}{4\pi}\,\frac{Ze(-\mathbf{v}_e \times \mathbf{r})}{r^3} = \frac{\mu_0}{4\pi}\,\frac{Ze(\mathbf{r} \times \mathbf{v}_e)}{r^3}\,.$$

4. Expressing in terms of orbital angular momentum

The orbital angular momentum of the electron is

$$\mathbf{L} = m_e\,\mathbf{r} \times \mathbf{v}_e\,.$$

Substituting into the expression for $\mathbf{B}_{\text{eff}}$,

$$\mathbf{B}_{\text{eff}} = \frac{\mu_0}{4\pi}\,\frac{Ze}{m_e r^3}\,\mathbf{L}\,.$$

Spin-Orbit Interaction

Electron moves around the nucleus in an orbital plane

$\mathbf{L}$ is perpendicular to the orbital plane

In the electron's rest frame, the positive nucleus appears to circulate

A circulating positive charge behaves like a current loop

$\mathbf{B}_{\text{eff}}$ is perpendicular to the orbital plane

$\mathbf{B}_{\text{eff}} \parallel \mathbf{L}$

Spin-Orbit Interaction

But the electron already possesses

$$\boldsymbol{\mu}_S \propto \mathbf{S}\,.$$

Therefore (similar to Zeeman Effect but not due to external magnetic field),

$$U = -\boldsymbol{\mu}_S \cdot \mathbf{B}_{\text{eff}} = \boldsymbol{L}\,.\boldsymbol{S} = \Delta H_{\text{SO}}$$

Spin-Orbit Interaction

Suppose L and S make an angle θ

$$\mathbf{L} \cdot \mathbf{S} = LS\cos\theta\,.$$

Therefore,

$$\Delta E_{\mathrm{SO}} \propto LS\cos\theta\,.$$

So the spin–orbit energy depends on the **relative orientation of the orbital and spin angular momenta**.

Schematically,

$$\mathbf{L}, \mathbf{S} \text{ more aligned} \quad \Rightarrow \quad \mathbf{L} \cdot \mathbf{S} > 0,$$

$$\mathbf{L}, \mathbf{S} \text{ more opposed} \quad \Rightarrow \quad \mathbf{L} \cdot \mathbf{S} < 0.$$

Hence the Hamiltonian becomes

$$\hat{H} = \hat{H}_0 + \hat{H}_{\mathrm{SO}},$$

with

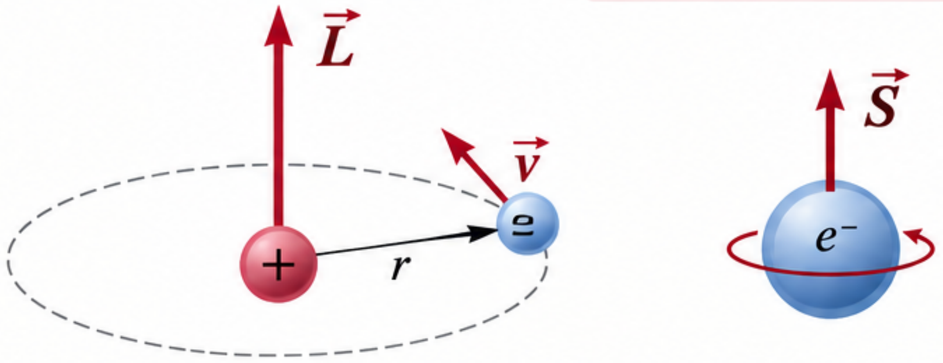
$$\hat{H}_{\mathrm{SO}} = \xi(r)\,\mathbf{L} \cdot \mathbf{S}\,.$$

Orbital motion + spin: total angular momentum

- **The electron has two angular momenta**

Orbital angular momentum $\vec{L}$
(orbital motion around nucleus)

Spin angular momentum $\vec{S}$
(intrinsic spin of the electron)



- **The electron experiences a spin–orbit interaction**

$$H_{SO} = \xi(r)\,\vec{L}\cdot\vec{S}$$

$\vec{L}$ and $\vec{S}$ are not independent — they interact with each other.

If both $\vec{L}$ and $\vec{S}$ are part of the electron's motion and interact with each other, what angular momentum describes the atom as a whole?

$$\vec{J} = \vec{L} + \vec{S}$$

Total angular momentum

1 Relation between $\vec{L}\cdot\vec{S}$ and $\vec{J}$

From $\vec{J} = \vec{L} + \vec{S}$,

$$J^2 = (L+\vec{S})^2$$
$$= L^2 + S^2 + 2\vec{L}\cdot\vec{S}$$

$$\vec{L}\cdot\vec{S} = \frac{1}{2}\left(J^2 - L^2 - S^2\right)$$

Therefore, the spin–orbit interaction is naturally described in terms of the total angular momentum J.

2 Quantum numbers

J^2 has eigenvalues

$$J^2 \;\rightarrow\; j(j+1)\hbar^2$$

with

$$j = |l-s|,\ |l-s|+1,\ \ldots,\ l+s$$

The projection along the z-axis is

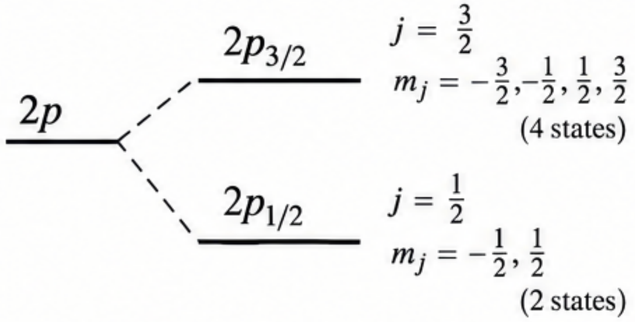
$$J_z \;\rightarrow\; m_j\hbar$$
$$m_j = -j, -j+1, \ldots, +j$$

j is the magnitude quantum number.
m_j runs from $-j$ to $+j$.

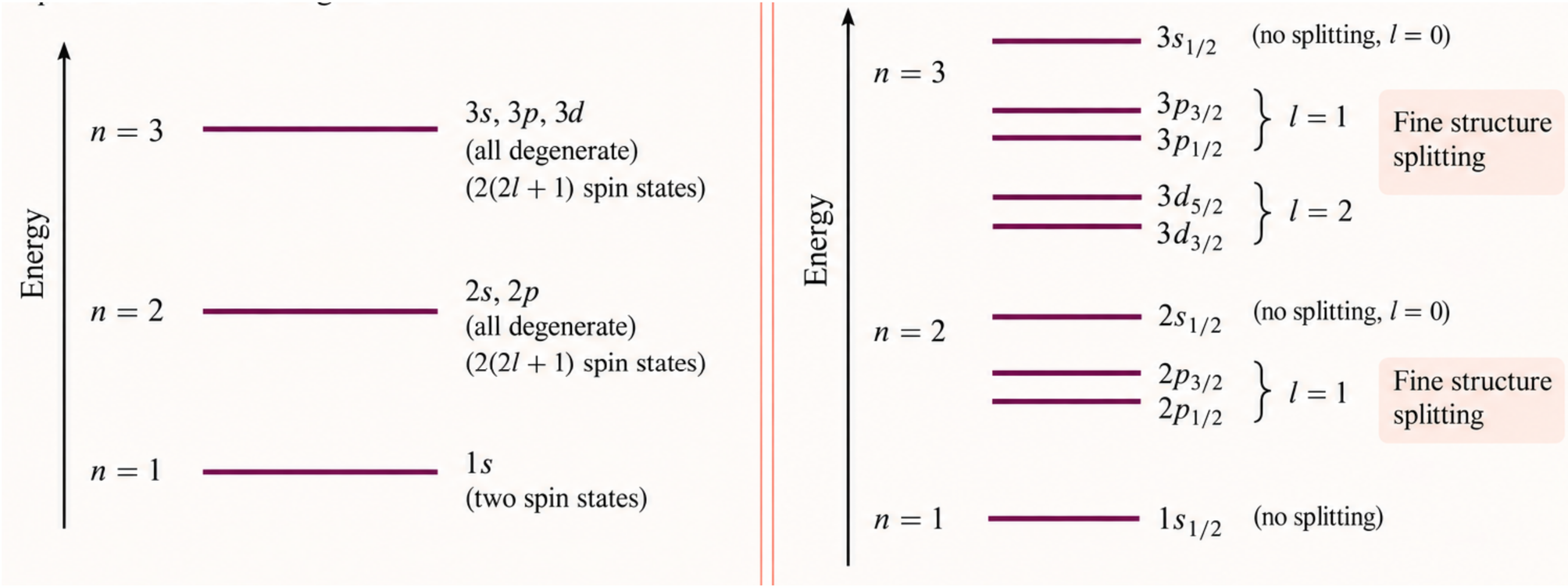
3 Example: 2p level

For $2p$, $l = 1$ and $s = \frac{1}{2}$. Therefore,

$$j = \frac{1}{2},\ \frac{3}{2}$$



Spin-Orbit Interaction



X-Ray spectra

We now know that the state of an electron in an atom is described by quantum numbers, and spin–orbit coupling gives a finer structure to the atomic energy levels.

What happens if an energetic electron strikes an atom and removes an inner-shell electron?

$$\text{Energetic } e^- + \text{Atom} \longrightarrow \text{inner-shell vacancy}$$

The ejected electron becomes free, but the atom is now left in an excited state with a vacancy.

X-Ray spectra

What happens to the inner-shell vacancy?

Higher-shell electron $\longrightarrow$ lower-shell vacancy

and

The lost atomic energy is emitted as a photon.

$$E_\gamma = E_i - E_f\,.$$

We can classify:

$$L \to K \quad \Rightarrow \quad K_\alpha$$
$$M \to K \quad \Rightarrow \quad K_\beta$$

Because these energies are determined by the atomic energy levels, the emitted X-ray energies are characteristic of the element.

X-Ray spectra

Does an $L \rightarrow K$ transition always produce an X-ray?

No.

It produces a photon whose energy is simply the difference between the two levels. Whether that photon is an X-ray depends on Z.

For a hydrogen-like atom,

$$E_n = -13.6 \frac{Z^2}{n^2} \text{ eV} .$$

In a many-electron atom, other electrons screen the nuclear charge, so qualitatively we replace Z by something like

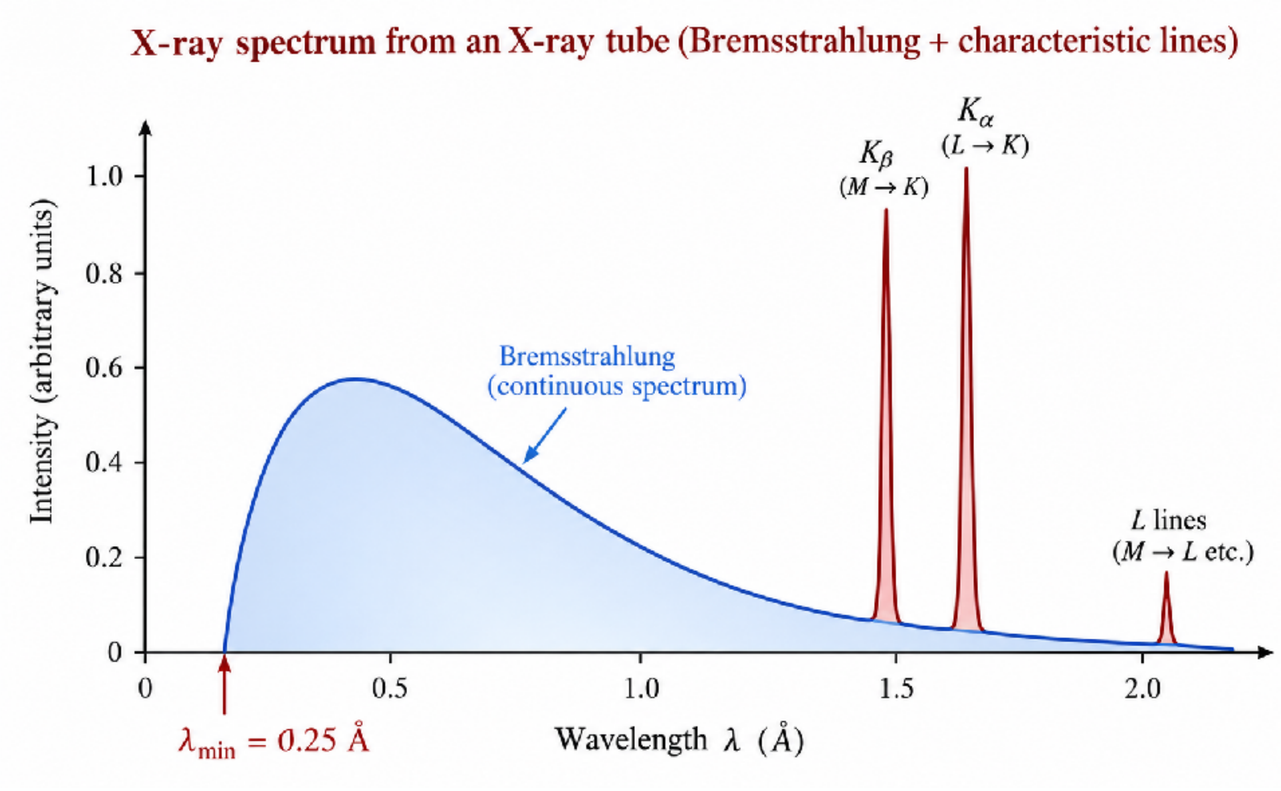
$$Z_{\text{eff}} < Z .$$

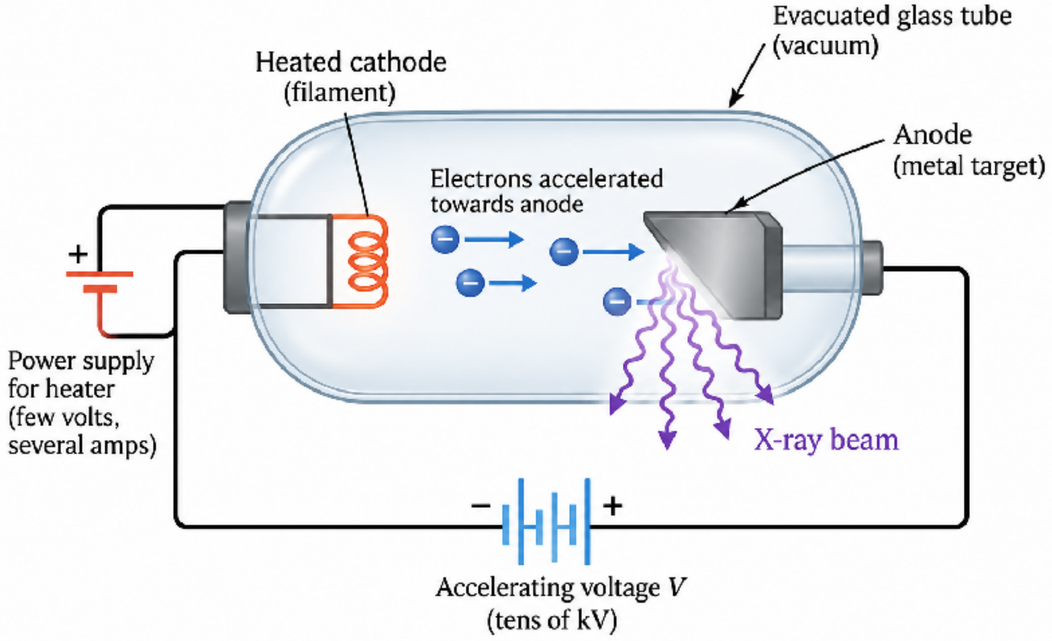
X-Ray spectra

So when energetic electrons bombarded the metal anode, two different mechanisms were operating:

electron deceleration ⟶ Bremsstrahlung continuum

inner-shell atomic transitions ⟶ characteristic X-ray lines





X-Ray spectra

But we now know the levels have finer structure

Start with a question:

Is 2p really one single energy level?

No.

Spin–orbit coupling gives

$$2p \longrightarrow 2p_{1/2}, \qquad 2p_{3/2}\,.$$

Therefore both transitions are possible:

$$2p_{3/2} \rightarrow 1s_{1/2} \qquad \Rightarrow \qquad K_{\alpha 1}$$

$$2p_{1/2} \rightarrow 1s_{1/2} \qquad \Rightarrow \qquad K_{\alpha 2}$$

Since

$$E(2p_{3/2}) \neq E(2p_{1/2}),$$

the emitted photons have slightly different energies:

$$E_{K_{\alpha 1}} \neq E_{K_{\alpha 2}}\,.$$